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Bullet seating die

Abstract

A bullet seating die for use with an ammunition press to seat a bullet in a case includes a die housing having an ammunition press mount for mounting the bullet seating die on the ammunition press. The die housing defines a bullet receiving space having an open end and a side window in communication with the bullet receiving space. The side window sized and shaped to permit the bullet to be inserted therethrough to position the bullet in the bullet receiving space. A seating stem is supported by the die housing. The seating stem is arranged to engage the bullet to press the bullet into the case when the ammunition press is operated. A spring biases the seating stem away from the open end of the bullet receiving space to permit the bullet to be inserted through the side window and into the bullet receiving space.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/444,256, filed Aug. 2, 2021, and claims priority to U.S. Provisional Application No. 63/059,398, filed Jul. 31, 2020, the entirety of each of which is hereby incorporated by reference.

FIELD

(1) The present disclosure generally relates to ammunition loading equipment, and more particularly to a bullet seating die.

BACKGROUND

(2) Referring to FIG. 1, a bullet 1 can be seated into a charged case or cartridge 2 using an ammunition press, such as the ammunition press 3. A charged case 2 is a case that has been resized and primed, has powered placed into it and is ready to accept a bullet. The ammunition press 3 includes a threaded receiver at the top into which a conventional seating die 4 is threadably attached. Typically, conventional seating dies are caliber and case specific, meaning each seating die can only be used with one caliber size and one type/size of case. To seat the bullet 1 in the case 2, the case is placed on the ammunition press and then the bullet is placed on the top of the case. The ammunition press 1 is then operated to move the case 2 and bullet 1 upward into the seating die 4. As the bullet 1 is raised into the seating die 4, the bullet engages the seating die, preventing further upward movement and thereby pressing the bullet into the case 2 as the case continues to be moved upward by the ammunition press 3. After the bullet 1 is seated in the case 2, the user resets the ammunition press and can repeat the process for another bullet and case.

SUMMARY

(3) In one aspect, a bullet seating die for use with an ammunition press to seat a bullet in a case comprises a die housing having an ammunition press mount configured to mount the bullet seating die on the ammunition press. The die housing defines a bullet receiving space having an open end. The die housing defines a side window in communication with the bullet receiving space and sized and shaped to permit the bullet to be inserted therethrough to position the bullet in the bullet receiving space. A seating stem is supported by the die housing within the bullet receiving space. The seating stem is arranged to engage the bullet to press the bullet into the case when the ammunition press is operated. A spring biases the seating stem away from the open end of the bullet receiving space so that the seating stem is positioned relative to the side window to permit the bullet to be inserted through the side window and into the bullet receiving space.

(4) In another aspect, a method of seating a bullet in a case using an ammunition press comprises placing the bullet in a bullet receiving space of a bullet seating die mounted to the ammunition

press and operating the ammunition press to seat the bullet in the case. The placing includes inserting the bullet through a side window of the bullet seating die under a seating stem biased away from a lower end of the side window. The seating stem does not need to be moved to place the bullet into the bullet receiving space.

(5) Other objects and features of the present disclosure will be in part apparent and in part pointed out herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an illustration of an ammunition press with a conventional seating die;
- (2) FIG. 2 is a perspective of a bullet seating die according to one embodiment of the present disclosure;
- (3) FIG. 3 is a cross-section of the bullet seating die;
- (4) FIG. 4 is an exploded view of the bullet seating die;
- (5) FIG. 5 is a front perspective of the bullet seating die;
- (6) FIG. 6 is a front perspective of the bullet seating die showing the position of a seating stem when not raised by a spring;
- (7) FIG. 7 is a cross-section of an alignment sleeve of the bullet seating die;
- (8) FIG. 8 is a perspective of a bullet seating die kit according to one embodiment of the present disclosure within a case;
- (9) FIG. 9 is perspective of the bullet seating die kit; and
- (10) FIGS. 10A-C are side views of different sized seating stems of the bullet seating die kit.
- (11) Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

(12) Referring to FIGS. 2-4, one embodiment of a bullet seating die of the present disclosure is generally indicated by reference numeral **10**. The bullet seating die **10** is used with an ammunition press, such as the ammunition press **3** of FIG. 1, to seat a bullet **1** in a case **2**. The bullet seating die **10** includes a die housing **12**. The die housing **12** includes an ammunition press mount **14** configured to mount the bullet seating die **10** to the ammunition press **3**. In the illustrated embodiment, the ammunition press mount **14** comprises external threads **16** for threadably coupling the die housing **12** to the ammunition press **3**. The bullet seating die **10** also includes a lock nut or ring **18** threadably mountable on the threads **16**. After the die housing **12** is threadably mounted on the ammunition press **3**, the lock nut **18** can be used to secure the bullet seating die **10** to the ammunition press by engaging the lock nut against the ammunition press. Other ways of mounting the bullet seating die **10** on the ammunition press **3** are within the scope of the present disclosure.

(13) The die housing **12** is generally elongate and has opposite first and second ends. The die housing **12** defines an elongate or longitudinal bore **20** having open ends at each end of the die housing. As will become apparent, the longitudinal bore **20** receives various different components of the bullet seating die **10** and the bullet **1**. In particular, the die housing **12** defines (broadly, at least partially defines) a bullet receiving space **22**. The bullet receiving space **22** is a portion of the longitudinal bore **20**. The bullet receiving space **22** has an open end (e.g., an open lower end). The case **2** moves through the open end and into the bullet receiving space **22** when the bullet **1** is seated in the case. The open end of the bullet receiving space **22** is generally coextensive with one of the open ends (e.g., the lower end) of the bore **20**. In the illustrated embodiment, the die housing **12** includes a generally cylindrical side wall **24** defining the bore **20**. The die housing **12** also defines a side window **26**. The side window **26** is in communication with the bullet receiving space **22** (broadly, the longitudinal bore **20**). The side window **26** is sized and shaped to permit the bullet

1 to be inserted therethrough to position the bullet in the bullet receiving space **22**. This allows the bullet **1** to be inserted laterally into die housing **12**, making it easier to position the bullet **1** for seating with the case **2** over conventional bullet seating dies. For example, the conventional seating die **4** shown in FIG. **1** requires an operator to balance the bullet **2** on top of the case **1** to position the bullet for seating, which can be difficult.

(14) The bullet seating die **10** includes a seating stem **28** supported by the die housing **12**. The seating stem **28** is generally elongate and cylindrical. The seating stem **28** (e.g., an end thereof) is arranged to engage the bullet **1** to press the bullet into the case **2** when the ammunition press **3** is operated. The seating stem **28** resists the upward movement of the bullet **1** to force the bullet into the case **2** when the ammunition press is operated. The end (e.g., lower end) of the seating stem **28** that engages the bullet **1** can define a recess sized and shaped to receive the tip of the bullet. The seating stem **28** is disposed in the bullet receiving space **22** (broadly, the bore **20**). The seating stem **28** is positioned relative to the side window **26** in the die housing **12** such that the seating stem does not block or otherwise inhibit the bullet **1** from being inserted into the bullet receiving space **22**. In the illustrated embodiment, the bullet seating die **10** includes a spring **30** that positions the seating stem **28**. The spring **30** biases the seating stem **28** away from the open end of the bullet receiving space **22** (e.g., away from a lower end of the side window **26**) so that the seating stem is positioned relative to the side window to permit the bullet to be inserted through the side window and into the bullet receiving space **22**. The spring **30** holds the seating stem **28** up to keep the seating stem out of the way of the bullet **1**. The spring **30** also allows the position of the seating stem **28** to be adjusted, as described in more detail below.

(15) The seating stem **28** is positioned, via the spring **30**, to prevent the seating stem from blocking the side window **26** and inhibiting the bullet **1** from being inserted into the bullet receiving space **22**. Specifically, the seating stem **28** is positioned within the die housing **12** such that the lower end of the seating stem is spaced apart by a sufficient distance from the lower end of the side window **26** to permit the bullet **1** to be inserted therebetween to position the bullet in the bullet receiving space **22**. FIGS. **2** and **5** show the seating stem **28** positioned to create sufficient clearance to permit the bullet **1** to be inserted through the side window **26** and into the bullet receiving space **22**. FIG. **6** shows a version of the bullet seating die **10** where the die lacks a spring **30** or other element positioning the seating stem **28** relative to the side window such that an insufficient amount of clearance exists for inserting the bullet **1** into the bullet receiving space. The seating stem positioning shown in FIG. **6** is similar to conventional bullet seating dies having a seating stem that is free floating (e.g., is free to move downward under the influence of gravity) such that a user has to manually raise the seating stem in order to be able to place the bullet **1** in the bullet receiving space through the side window. Accordingly, it is easier to place a bullet **1** into the bullet receiving space **22** of the bullet seating die **10** of the present disclosure over corresponding conventional seating dies because the user does not need to move (e.g., lift) the seating stem **28** out of the way to create enough clearance to insert the bullet into the bullet receiving space.

(16) Referring to FIGS. **3** and **4**, the spring **30** is operatively disposed between the seating stem **28** and the die housing **12**. The bullet seating die **10** includes a first spring retainer **32** and a second spring retainer **34**. The spring **30** is disposed between and engages the first and second spring retainers **32**, **34**. While the spring **30** is shown in an uncompressed state in FIG. **3**, it is understood that the spring would be compressed between the first and second spring retainers **32**, **34** in this figure. The first spring retainer **32** is supported by the die housing **12**. The first spring retainer **32** has a generally disk shaped body **36** sized and shaped to be inserted into the bore **20** of the die housing. The die housing **12** includes a shoulder **38** within the bore **20**. The first spring retainer **32** (e.g., the disk shaped body **36**) is disposed in the bore **20** and engages (e.g., rests on) the shoulder **38**. The illustrated shoulder **38** is very narrow, such as about 0.5 mm. The first spring retainer **32** (e.g., a lower surface thereof) defines a closed end (e.g., closed upper end) of the bullet receiving space **22**. The disk shaped body **36** of the first spring retainer **32** defines an opening **40** (e.g., a

seating stem opening). The seating stem **28** extends through the opening **40** of the first spring retainer **32**. The seating stem **28** is slideable within the opening **40** relative to the first spring retainer **32** to permit the position of the seating stem to be adjusted, as described in more detail below. One end (e.g., a first end) of the spring **30** engages the first spring retainer **32**. In the illustrated embodiment, the disk shaped body **36** defines a spring recess **42**. The spring recess **42** is aligned with the opening **40** and is sized and shaped to receive the end of the spring **30**. The second spring retainer **34** is supported by the seating stem **28**. In the illustrated embodiment, the second spring retainer **34** comprises an E-clip or a C-clip **44** mounted on the seating stem **28**. The seating stem **28** includes a circumferential channel **46** (FIGS. **10A-C**) that receives the E or C-clip **44**. Another end (e.g., a second end) of the spring **30** engages the second spring retainer **32**. In the illustrated embodiment, the spring **30** comprises a coil spring. The seating stem **28** extends through the coil spring.

(17) Referring to FIGS. **2-4**, the bullet seating die **10** includes a cap **48** supported by the die housing **12**. In particular, the cap **48** is mounted on an end of the die housing **12**. The cap **48** includes a seat **50** (FIG. **3**). The seat **50** is arranged to engage the seating stem **28** (e.g., an end thereof) to set the position of the seating stem relative to the die housing **12**. The spring **30** biases the seating stem **28** against the seat **50**. In the illustrated embodiment, the seat **50** defines a recess **52** sized and shaped to receive the end of the seating stem **28**. The cap **48** includes an adjuster **54**. The adjuster is configured to move the seat **50** relative to the die housing **12** to set the position of the seating stem **28** relative to the die housing. This allows the overall depth the bullet **1** is seated in the case **2** to be adjusted. Moving the seat **50** toward the open end of the bullet receiving space **22** (e.g. in one direction) compresses the spring **30** and moves the seating stem **28** toward the open end of the bullet receiving space. This increases the depth the bullet **1** is seated into the case **2**. Moving the seat **50** away from the open end of the bullet receiving space **22** (e.g., in the opposite direction) permits the spring **30** to expand and moves the seating stem **28** away from the open end of the bullet receiving space. This decreases the depth the bullet **1** is seated into the case **2**. In the illustrated embodiment, the adjuster **54** comprises a micrometer **56**. The micrometer **56** is threadably connected to the die housing **12**. The die housing **12** includes external threads that engage internal threads of the micrometer **56**. The micrometer **56** is operatively coupled to the seat **50** with a fastener **58** such that the seat **30** moves longitudinally with the micrometer as the micrometer moves longitudinally along the die housing **12** when the micrometer is rotated. Rotating the micrometer **56** in one direction (e.g., a clockwise direction) moves the seat **50** and seating stem **28** toward the open end of the bullet receiving space **22** and rotating the micrometer in the opposite direction (e.g., a counter-clockwise direction) moves the seat and seating stem away from the open end of the bullet receiving space. Desirably, micrometer **56** enables precise (e.g., about 0.001 inch (about 0.025 mm)) adjustment to the seating depth of the bullet **1**.

(18) Referring to FIGS. **3** and **7**, the bullet seating die **10** includes an alignment sleeve **60** (a bullet holder). The alignment sleeve **60** is configured to guide the bullet **1** into the case **2** when the ammunition press **3** is operated. The alignment sleeve **60** is sized and shaped to be disposed within the bullet receiving space **22** and to slide within the bullet receiving space. The alignment sleeve **60** is generally cylindrical. The alignment sleeve **60** includes an exterior shoulder **62** that engages an interior shoulder **64** of the die housing **12** in the bore **20** to hold the alignment sleeve within the bullet receiving space **22** and prevent the alignment sleeve from sliding out of the open end of the bullet receiving space. The alignment sleeve **60** defines an alignment lumen **62** therethrough. The alignment lumen **62** includes a bullet guide section **62A**, a bullet alignment section **62B**, a case alignment section **62C** and a case guide section **62D**. The bullet guide section **62A**, the bullet alignment section **62B**, the case alignment section **62C** and the case guide section **62D** are arranged end to end from one end of the alignment sleeve **60** to the other. The bullet guide section **62A** has an inverted conical shape and guides the bullet **1** into the bullet alignment section **62B** when the bullet is inserted into the bullet receiving space **22**. The case guide section **62D** has a conical shape

and guides the case **2** into the case alignment section **62C** when the case is raised up into the bullet receiving die **10**. The bullet alignment section **62B** is arranged relative to the case alignment section **62C** such that the bullet alignment section aligns and guides the bullet **1** into the case **2** when the bullet is seated in the case. The alignment lumen **62** is sized and shaped to seat a specific caliber bullet **1** into its corresponding specific caliber case **2**. The bullet alignment section **62B** has a diameter to match a specific caliber bullet **1**, such as a .223 caliber bullet. Likewise, the case alignment section **62C** has a diameter to match a specific caliber case **2** (same caliber as the bullet alignment section **62B**), such as a .223 caliber case. The case alignment section **62C** is sized and shaped (e.g., has a height) that allows the case alignment section to receive any case **2** of the caliber corresponding to the caliber of the case alignment section **62C**. The case alignment section **62C** is also sized to receive any wall thickness of a case **2**, as a neck of the case (the portion that accepts the bullet **1**) can vary between cases (even the same caliber cases). This allows the bullet seating die **10** (specifically, the alignment sleeve **60**) to be used with any case **2** of a specific caliber (e.g., any .223 caliber case). It is understood, the alignment lumen **62** can be sized and shaped for generally any caliber such as .223, .243, .257, .264, .277, .284, .308, .325, .338, etc.

(19) The alignment sleeve **60** includes a bullet retainer **64**. The bullet retainer **64** is arranged to engage the bullet **1** (e.g., a lower end thereof) to hold the bullet within the bullet receiving space **22** when the bullet is placed in the bullet receiving space. Without the bullet retainer **64**, the bullet **1** may fall out of the bullet seating die **10** (e.g., fall through the alignment lumen **62**). By holding the bullet **1** in the bullet receiving space **22** with the bullet retainer **64**, the user does not have to balance the bullet on top of a case **2** when operating the ammunition press. In the illustrated embodiment, the bullet retainer **64** comprises a cylindrical rod **66**. In another embodiment, the bullet retainer may comprise a spring clip. The bullet retainer **64** is resiliently biased by a bullet retainer spring **68**. The bullet retainer spring **68** biases the bullet retainer **64** into a retaining position (FIG. 7), where at least a portion of the bullet is disposed in the bullet alignment section **62B** to engage the lower end of the bullet **1** to hold the bullet and prevent the bullet from falling through the bullet alignment section. In the illustrated embodiment, the rod **66** is slidably disposed within a channel **70**. In operation, the bullet retainer **64** is moved out of the bullet alignment section **62B** by the bullet **1**, away from the retaining position, when the bullet is pressed by the seating stem **28**. Specifically, the bullet **1** moves the rod **66** outward along the channel **70** and away from the retaining position when the bullet is pressed by the seating stem **28** to move the lower end of the bullet past the bullet retainer **64** when the ammunition press **3** is operated. In the illustrated embodiment, the bullet retainer spring **68** comprises an O-ring **72**. The O-ring **72** engages the opposite ends of the rod **66** to bias the rod toward the retaining position.

(20) Referring to FIGS. 8-10C, the bullet seating die **10** may be part of a bullet seating die kit **100**. The bullet seating die kit **100** includes the bullet seating die **10**, a plurality of different alignment sleeves **60**, and a plurality of different seating stems **28**. The plurality of alignment sleeves **60** are all interchangeable with one another in the bullet seating die **10**. Likewise, the plurality of seating stems **28** are also all interchangeable with one another in the bullet seating die **10**. In the illustrated embodiment, the bullet seating die kit **100** includes a .224 caliber alignment sleeve **60A**, a .243 caliber alignment sleeve **60B**, a .257 caliber alignment sleeve **60C**, a .264 caliber alignment sleeve **60D**, a .277 caliber alignment sleeve **60E**, a .284 caliber alignment sleeve **60F**, a .308 caliber alignment sleeve **60G** (shown in the bullet receiving space **22** of the die housing **12**), a .325 caliber alignment sleeve **60H**, and a .338 caliber alignment sleeve **60I**. More or fewer alignment sleeves and/or other caliber alignment sleeves are within the scope of the present disclosure. By selecting (e.g., interchanging) a desired caliber alignment sleeve **60A-I**, the bullet seating die **10** can be configured for a specific caliber of bullet **1**. In addition, in the illustrated embodiment, the bullet seating die kit **100** includes a short-length seating stem **28A** (shown mounted in the die housing **12**), a medium-length seating stem **28B**, and a long-length seating stem **28C**. More or fewer seating stems and/or other lengths of seating stems are within the scope of the present disclosure. Bullets **1**

and cases **2** come in different lengths (even for the same caliber). The different length seating stems **28A-C** enable the bullet seating die **10** to be configured to match the length of bullet **1** and/or case **2** being seated together in order to properly set the depth the bullet is seated in the case. For example, a 2.2 inch long case requires a different configuration than a 3.3 inch long case. By selecting a seating stem **28A-C** of a specific length, the desired seating depth can be achieved. It is understood that the adjuster **54** is used for more fine tune adjustments than interchanging seating stems **28A-C** of different lengths allows (e.g., selecting a seating stem of a specific length allows for large scale adjustments). The seating stems **28A-C** are configured to be used with multiple caliber bullets (e.g., the seating stems are not caliber specific). For example, the taper at the end of the medium-length and long-length seating stems **28B-C** enables these seating stems to be used with different (e.g., smaller) caliber alignment sleeves **60A-I** and their corresponding bullets **1**. The bullet seating die kit **100** may include a box or case **102** for housing the components (e.g., bullet seating die **10**, alignment sleeves **60A-I**, seating stems **28A-C**) of the kit.

(21) To change out the seating stem **28** and/or alignment sleeve **60** of the bullet seating die **10**, a user removes the cap **48** from the end of the die housing **12**. The user unscrews the adjuster **54** from the end of the die housing **12**. This opens up the end of the bore **20** and allows the seating stem **28**, the spring **30**, the first spring retainer **32**, and the alignment sleeve **60** to be removed from the bore. The seating stem **28**, the spring **30**, the first spring retainer **32**, and the alignment sleeve **60** can all slide out of the bore **20** through the end previously closed by the cap **48** (e.g., the end opposite the open end of the bullet receiving space **22**). After the seating stem **28** and the alignment sleeve **60** are removed from the die housing **12**, the user can interchange the existing seating stem and/or alignment sleeve with another (e.g., different size) seating stem and/or alignment sleeve, respectively, as desired to configure the bullet seating die **10** to the desired caliber and size of the bullet **1** and the case **2** that are being seated together. To reassemble the bullet seating die **10**, the user inserts the desired alignment sleeve **60** (e.g., one of alignment sleeves **60A-I**), case guide section **52D** first, into the bore **20** of the die housing **12**. The alignment sleeve **60** slides down the bore **20** and engages the shoulder **64** of the die housing **12**, positioning the alignment sleeve in the bullet receiving space **22**. After, the user inserts the first spring retainer **32** and then the spring **30** into the bore **20** of the die housing **12**. Then, the user inserts the desired seating stem **28** (e.g., one of seating stems **28A-C**) into the bore **20** of the die housing **12**, making sure the seating stem is aligned with the opening **40** in the first spring retainer so that the seating stem slides therein. Lastly, the user reattaches the cap **48** to the end of the die housing **12**. The user screws the adjuster **54** onto the end of the die housing **12**. The bullet seating die **10** can then be mounted onto the ammunition press (if not already).

(22) A method of seating a bullet **1** in a case **2** using the bullet seating die **10** will now be described. The user configures the bullet seating die **10** by selecting the appropriate alignment sleeve **60** and/or seating stem **28** for the bullet **1** and case **2** to be seated or loaded. The user installs the selected alignment sleeve **60** and/or seating stem **28** into the die housing **12** as described herein. After the bullet seating die **10** is assembled, the user mounts the bullet seating die **10** on the ammunition press **3**, as described herein. The bullet seating die **10** can then be adjusted to set the desired depth of the bullet **1** in the case **2**. The user actuates the adjuster **54** (e.g., rotates the micrometer **56**) to adjust the position of (e.g., move) the seating stem **28** relative to the die housing **12** to set the seating depth of the bullet **1** in the case **2**. Now the ammunition press **3** is ready to seat a bullet **1** in a case **2**. The user puts the case **2** in the ammunition press **3**. The user also places the bullet **1** in the bullet receiving space **22**. The bullet **1** is inserted through the side window **26** and into the bullet receiving space **22**. The user inserts the bullet **1** under the seating stem **28**, which is biased away from the lower end of the side window **26**. Because the seating stem **28** is held up or biased by the spring **30**, the user does not need to move the seating stem in order to place the bullet **1** into the bullet receiving space **22**. This makes placing the bullet **1** in the bullet receiving space **22** easier over conventional seating dies. When the bullet **1** is in the bullet receiving space **22**, the

bullet retainer **64** engages the bullet to hold the bullet in the bullet receiving space. With the bullet **1** positioned in the bullet seating die **10**, the user then operates the ammunition press **3**. The ammunition press **3** may include a lever the user turns to operate the ammunition press. When the ammunition press **3** is operated, the case **2** is raised by the press toward the bullet **1** in the bullet seating die **10**. As the case **2** is raised, the upper end of the case moves into the alignment sleeve **60** (e.g., the case alignment section **62C**). The case **2** engages the alignment sleeve **60** and pushes the alignment sleeve and bullet **1** upward within the bullet receiving space **22** toward the seating stem **28**. Eventually, the nose of bullet **1** engages the end (e.g., lower end) of the seating stem **28**, thereby preventing any further upward movement of the bullet. Continued upward movement of the case **2** and the alignment sleeve **60** by the ammunition press **3** seats the bullet **1** in the case. The engagement of the bullet **1** with the seating stem **28** prevents further upward movement of the bullet so that as the press continues to move the case **2** and alignment sleeve **60** upward, the bullet deflects the bullet retainer **64** and is pushed into the case. The ammunition press **3** is continued to be operated until the case **2** is raised to its upper most point, which seats the bullet **1** in the case at the seating depth. The ammunition press **3** is then operated to lower the case **2**. As the case **2** is lowered, the case brings the now seated bullet **1** with the case and the case and bullet withdraw from the alignment sleeve **60**. The alignment sleeve **60** rests at the bottom of the die housing **12**, ready to receive the next bullet from the side window **26**. The user removes the seated bullet **1** and case **2** from the ammunition press **3** and then repeats the process with the next bullet and case.

(23) When introducing elements of the present disclosure or the preferred embodiments(s) thereof, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

(24) It will be apparent that modifications and variations are possible without departing from the scope defined in the appended claims.

(25) As various changes could be made in the above constructions and methods without departing from the scope of the disclosure, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A bullet seating die kit for use with an ammunition press to seat a bullet in a case, the bullet having a first caliber or a second caliber different than the first caliber, the bullet seating die kit comprising: a die housing having an ammunition press mount configured to mount the bullet seating die on the ammunition press, the die housing defining a bullet receiving space having an open end, the die housing defining a side window in communication with the bullet receiving space and sized and shaped to permit the bullet to be inserted therethrough to position the bullet in the bullet receiving space; and a seating stem configured to be supported by the die housing within the bullet receiving space, the seating stem being arranged to engage the bullet to press the bullet into the case when the seating stem is supported by the die housing and the ammunition press is operated; and first and second bullet holders, the first bullet holder configured to hold the bullet when the bullet is of the first caliber and the second bullet holder configured to hold the bullet when the bullet is of the second caliber, the first and second bullet holders being configured to be interchangeably receivable in the bullet receiving space of the die housing, the first bullet holder being interchangeably receivable in the bullet receiving space to configure the bullet seating die kit to seat the bullet when the bullet is of the first caliber, and the second bullet holder being interchangeably receivable in the bullet receiving space to configure the bullet seating die kit to seat the bullet when the bullet is of the second caliber.

2. The bullet seating die kit of claim 1, further comprising a cap configured to be supported by the die housing, the cap including a seat configured to engage the seating stem when the seating stem

and the cap are supported by the die housing.

3. The bullet seating die kit of claim 2, wherein the cap includes an adjuster configured to move the seat relative to the die housing to set the position of the seating stem relative to the die housing when the seating stem and the cap are supported by the die housing.

4. The bullet seating die kit of claim 3, wherein the adjuster comprises a micrometer configured to threadably connected to the die housing.

5. The bullet seating die kit of claim 1, wherein the first bullet holder includes a first bullet retainer arranged to engage the bullet to hold the bullet within the bullet receiving space when the first bullet holder is received in the bullet receiving space, and wherein the second bullet holder includes a second bullet retainer arranged to engage the bullet to hold the bullet within the bullet receiving space when the first bullet holder is received in the bullet receiving space.

6. The bullet seating die kit of claim 5, wherein the first bullet retainer is resiliently biased by a first bullet retainer spring and the second bullet retainer is resiliently biased by a second bullet retainer spring.

7. The bullet seating die kit of claim 6, wherein the first bullet retainer spring comprises a first O-ring and the second bullet retainer spring comprises a second O-ring.

8. The bullet seating die kit of claim 1, wherein the first bullet holder is configured to guide the bullet into the case when first bullet holder is received in the bullet receiving space and the ammunition press is operated, and wherein the second bullet holder is configured to guide the bullet into the case when the second bullet holder is received in the bullet receiving space and the ammunition press is operated.

9. The bullet seating die kit of claim 1, wherein the first bullet holder is moveable within the bullet receiving space when the first bullet holder is received in the bullet receiving space, and wherein the second bullet holder is moveable within the bullet receiving space when the second bullet holder is received in the bullet receiving space.

10. The bullet seating die kit of claim 1, further comprising a third bullet holder configured to hold the bullet when the bullet is of a third caliber different from the first and second calibers, the third bullet holder being interchangeably receivable in the bullet receiving space to configure the bullet seating die kit to seat the bullet when the bullet is of the third caliber.

11. The bullet seating die kit of claim 10, further comprising a fourth bullet holder configured to hold the bullet when the bullet is of a fourth caliber different from the first, second, and third calibers, the fourth bullet holder being interchangeably receivable in the bullet receiving space to configure the bullet seating die kit to seat the bullet when the bullet is of the fourth caliber.

12. The bullet seating die kit of claim 8, wherein the first bullet holder is configured to hold the bullet when the first caliber is a .223 caliber and the second bullet holder is configured to hold the bullet when the second caliber is a .308 caliber.

13. The bullet seating die kit of claim 10, wherein the first bullet holder is configured to hold the bullet when the first caliber is a .223 caliber, the second bullet holder being configured to hold the bullet when the second caliber is a .308 caliber, and the third bullet holder being configured to hold the bullet when the third caliber is a 30-06 caliber.

14. The bullet seating die kit of claim 1, further comprising a spring configured to bias the seating stem away from the open end of the bullet receiving space so that the seating stem is positioned relative to the side window to permit the bullet to be inserted through the side window and into the bullet receiving space.

15. The bullet seating die kit of claim 14, wherein the spring is configured to be operatively disposed between the seating stem and the die housing.

16. The bullet seating die kit of claim 15, further comprising a first spring retainer configured to be supported by the die housing, a first end of the spring being configured to engage the first spring retainer.

17. The bullet seating die kit of claim 16, further comprising a second spring retainer configured to

be supported by the seating stem, a second end of the spring being configured to engage the second spring retainer.

18. The bullet seating die kit of claim 17, wherein the spring comprises a coil spring, the seating stem is configured to extend through the coil spring.

19. The bullet seating die kit of claim 18, wherein the die housing defines an elongate bore and includes a shoulder within the elongate bore, the first spring retainer being configured to be disposed in the elongate bore and engage the shoulder.

20. The bullet seating die kit of claim 19, wherein the second spring retainer comprises an E-clip configured to mount on the seating stem.

21. The bullet seating die kit of claim 19, wherein the first spring retainer is configured to define an end of the bullet receiving space.

22. The bullet seating die kit of claim 16, wherein the first spring retainer defines an opening, the seating stem being configured to extend through the opening and be slidable within the opening relative to the first spring retainer.
